THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 08

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** 81	$9^2 = 81$
2	**(a)** 729	$9 \times 9 \times 9 = 729$
3	**(c)** 7^6	Add the exponents: $3 + 3 = 6$
4	**(a)** 11001	Repeated division of 25 by 2
5	**(b)** rectangle	Rectangle property
6	**(a)** 110°	Opposite angles are equal
7	**(c)** 2	Digit sum must be a multiple of 3
8	**(d)** non-negative	A square is never negative
9	**(b)** ₹ 192	₹ 24 each
10	**(b)** becomes half	Product constant
11	**(a)** 540	450 × 120/100
12	**(a)** 62.5%	Multiply by 100
13	**(d)** Yes	$14^2 + 48^2 = 50^2$
14	**(c)** 25 cm	$\sqrt{7^2 + 24^2} = 25$
15	**(c)** 36°	Angle = percent × 3.6
16	**(b)** 12	Euler's relation
17	**(b)** a rapid increase	Slope
18	**(b)** 36	$4x + 16$
19	**(c)** 70 cm²	$\frac{1}{2} \times b \times h$
20	**(c)** 10 cm	Area ÷ length

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$a^n \div a^n = 1$ and by the law also $a^{n-n} = a^0$ (2).	2
22	60 minutes (1); our time measurement still uses 60 (1).	1+1
23	$A = 5$ (2).	2
24	1 unit on the map is 50000 of the same units on the ground (2).	2
25	PR (2).	2
26	Direct: ratio constant; inverse: product constant (2).	2
27	Both 50 (2).	2
28	5 : 8 (2).	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	$(80 - 2)^2 = 6400 - 320 + 4$ (2); $= 6084$ (1).	2+1
30	$\frac{1}{2}(10 + 6) \times 4$ (2); 32 cm² (1).	2+1
31	$9m^2 - 12mn + 4n^2$ (2); $36 - 24 + 4 = 16$ (1).	2+1
32	$A = P(1 + r/100)^n = 8000 \times (1 + 5/100)^3 = ₹9,261.00$ (2); CI = ₹1,261.00 (1).	2+1
33	Start with an equilateral triangle (1); divide each side into three equal parts and add a small triangle on the middle part (1); repeat on every new side (1).	1+1+1
34	Choose equal or closest numbers (1); 9 and 9 (1); product 81 (1).	1+1+1
35	Diagonal 17 m (1.5); saved $23 - 17 = 6$ m (1.5).	1½+1½
	OR: By the theorem, height $= \sqrt{25^2 - 7^2}$ (2); $= 24$ m (1).	2+1

36	Trapezium $= \frac{1}{2}(9+15) \times 10 = 120$ cm² (2); rectangle 150 cm² (1).	2+1
	OR: Area 260 m² (1.5); perimeter 66 m, cost ₹ 1980 (1.5).	1½+1½

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) $a^{m-m} = a^0 = 1$ (3). (ii) $9 \times \frac{1}{9} = 1$ (2).	3+2
	OR: (i) Compare the powers of 10 first, then the decimal parts (3). (ii) $4.5 \times 10^7 = 45 \times 10^6 > 9.1 \times 10^6$ (2).	3+2
38	(i) ₹ 18000 (2). (ii) ₹ 68000 (1). (iii) ₹ 1888.89 (2).	2+1+2
	OR: (i) 24 girls, 6 with bicycles (3). (ii) 15% (2).	3+2
39	(i) Mean 8; deviations $-4, -2, 0, 2, 4$ sum 0 (3). (ii) Left and right deviations balance (2).	3+2
	OR: (i) $x = 8$ (2). (ii) 7.2 (3).	2+3
40	(i) 360 cm² (2). (ii) Foot distance 8 cm; area $\frac{1}{2} \times 8 \times 15 = 60$ cm² (3).	2+3
	OR: (i) Square 144, rectangle $16 \times 8 = 128$ (3). (ii) Square has the larger area (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND